

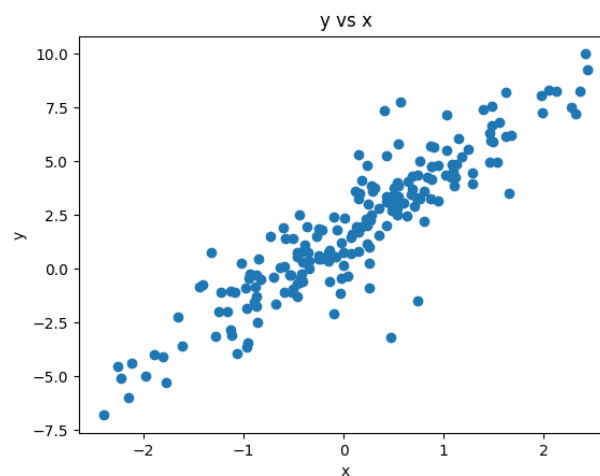
1 Reading the Shape of a Sample

Table 1: First Four Moments

Moment	Value
Mean	0.0011043942883968638
Variance	9.624822112193506e-05
Skewness	-0.6688314666632971
Kurtosis	5.339848648336579

- Based on the skewness of -0.669 and excess kurtosis of $5.339848648336579 - 3 = 2.339848648336579$, the Student's t, symmetric NIG, Normal inverse Gaussian, skewed t, and generalized hyperbolic distributions could have plausibly produced this sample. This is because the sample has positive excess kurtosis. Furthermore, skewness is close to 0 (satisfying Student's t and symmetric NIG) and not equal to 0 (satisfying Normal inverse Gaussian, skewed t, and generalized hyperbolic). Normal is filtered out of consideration because excess kurtosis is not close to 0. Lognormal is filtered out of consideration because the skew of the sample is negative.
- After fitting a normal with $\mu = 0.0011043942883968638$ and $\sigma = 0.009810617774734427$, it was found that 26 of the observations fall below the Normal's 1% quantile. In expectation, there should be $1000 \times 0.01 = 10$.
- Looking at the results from part b, the fitted Normal is underestimating how often low values occur (we expected 10 observations below the 1% quantile but got 26). The high excess kurtosis points to a fatter (left based on the 1% calculation) tail. We can also tell from negative skewness that the distribution leans left (heavier left tail relative to the right one). The direction of the error is that the Normal understates downside risk, predicting 1% frequency for events that actually happen $\sim 2.6\%$ of the time.

2 A Regression Whose Errors Are Not Normal

Figure 1: Plot of y Against x

- a. From the scatter plot, it appears that there is a strong, roughly linear relationship between x and y . Most of the observations are closely clustered together around a trend line.

The error term in question is the vertical distance from each point to the underlying linear trend. The vertical spread above and below the line looks fairly symmetric. There appears to be no sign of a skewed distribution in error terms (tails at $x = -2$ and $x = 2$ look similarly clustered). This eliminates Normal inverse Gaussian, skewed t, generalized hyperbolic, and lognormal distributions. Furthermore, there seem to be some data points (e.g. (0.5, -3)) that are much farther than the others. This might suggest heavier than Normal tails, eliminating the normal distribution. However, further testing would better judge kurtosis and how fat the tails are. This leaves Student's t and symmetric NIG distributions as plausible candidates.

- b. Based on the section title ("A Regression Whose Errors Are Not Normal") and the justification given in part a) about potentially heavy tails, Assumption 7: the error term is normally distributed, is violated by the data.

Given what I recall from previous econometrics courses I've taken, the consequences of a non-normally distributed error to the estimated slope are inconsequential. It should still remain unbiased (i.e. won't have a tendency to neither increase nor decrease the magnitude of the slope). However, it will affect the standard error of that slope. Its formula is as follows:

$$s.e.\hat{\beta} = s\sqrt{\text{diag}((X'X)^{-1})} = \sqrt{\frac{s^2}{\sum(x_i - \bar{x})^2}}$$

where

$$s^2 = \frac{\epsilon'\epsilon}{n-p}$$

The numerator of the standard error is affected by a violation of assumption 7. Because the errors are heavy tailed, unusually large residuals occur more often than a Normal distribution would predict. Since s^2 is built from squared residuals, those occasional large errors would disproportionately inflate sample variance, making it a noisier, less stable estimate of the true error variance than it would be under Normal errors. Consequently, the standard error, which is s scaled by a constant, also becomes noisier and less reliable. With fat tails, the real variability of $\hat{\beta}$ would then tend to be larger than what the formula would give. Using that standard error, which assumes that the errors are following a normal distribution, would understate the true standard error with heavier tails. This means that when constructing confidence intervals, it would come across as being more narrow than it actually is.

Table 2: Estimates of α, β and Fitted Error Parameters for the Three Models

Model	α	SE(α)	β	SE(β)	Root MSE	ν
1. OLS	1.560740	0.096269	2.990757	0.096640	1.351401	—
2. MLE (Normal)	1.560740	—	2.990757	—	1.344627	—
3. MLE (Student's t)	1.547980	—	3.019175	—	0.935513	3.632191

Using AICc, we get the following results:

Normal AICc: 692.1445231721832

Student's t AICc: 665.3274871722376

So the preferred model is Student's t.

- c. The violation of assumption 7 moved the intercept and slope estimate by a little bit. OLS and MLE under a Normal error gave the same estimates, as expected. Comparing the models under Normal errors against the one with Student's t error, the intercept, α , changed from 1.56 to 1.55 and β from 2.99 to 3.02. There are minimal changes to the slope, which is close to what I predicted (no change). The differences are likely because OLS weights every observation's influence on the fit by the square

of its residual, giving outliers outsized leverage and thereby changing the magnitude of β , whereas the Student's t model's likelihood saturates for large residuals and effectively down-weights them.

- d. The rejected model, the one that uses Normally distributed errors, gets wrong the spread of residuals. We can see this in the model using Student's t having an MSE of 0.94 compared to the models using Normal having an MSE of 1.34-1.35. Every observation whose actual residual happens to be large gets penalized hard in the Normal's log-likelihood, dragging the total down, even though the fitted line itself is nearly the same as the t-model's. A Normal distribution assigns very low probability density to large residuals (its tails decay fast). The Student's t model, by having a heavier tail (via ν), assigns those same large residuals meaningfully higher probability, so its log-likelihood doesn't get affected the same way.

Table 3: Quantile Comparison

Quantile	Normal	Student's t
95%	2.211714598444586	2.053832738412794
99.5%	3.4635296298790554	4.620082078259344

- e. The Normal is wider at the 95% quantile but t is wider at 99.5%, as shown in Table 3. This is because of the t distribution's heavier tails/excess kurtosis. Given both distributions integrate to the same total probability mass, the t distribution's fatter tails must be compensated for with a thinner 90-95% range. As a result, the t distribution looks less extreme than the Normal at moderate confidence levels but far more extreme moving out into its tail.

If I were to set up a capital buffer, it'd be important to use the model with a t distribution. A risk manager who used the Normal model to size a 99.5% capital buffer would set it at 3.4635, but the data itself (favoured by AICc) says the true 99.5% loss threshold is 4.6201, about 33% higher. Relying on the Normal model would leave the buffer under-capitalized against tail events, even though the Normal model looks perfectly adequate, or even conservative, at the more central 95% level. The correct error distribution matters most in the tails rather than the centre. In this context, a misspecified model can look fine on average while badly understating extreme risk, which is exactly where the capital buffer is supposed to protect against loss.

3 Pearson Against Spearman

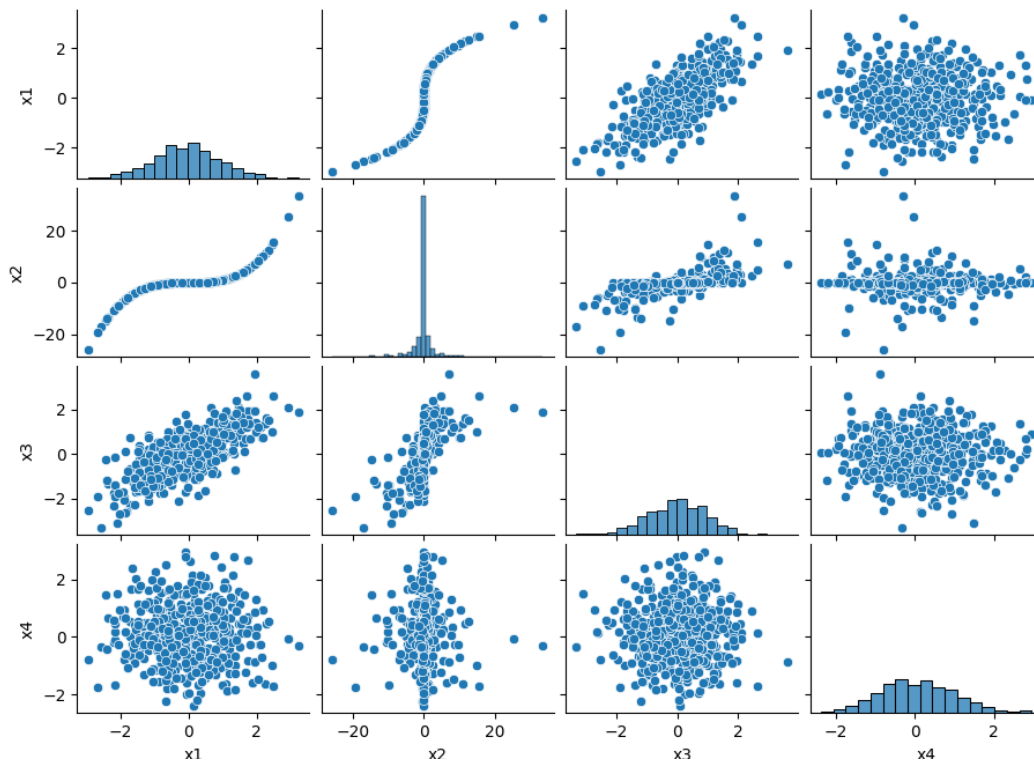


Figure 2: Plots of Every Pair in Series x1 through x4

- a. I expect the plots of x1-x2 and x2-x3 to have different coefficients. x1-x2 should have a strong Spearman coefficient because it's a monotonic function (weakly increasing). I would expect it to have a Spearman coefficient close to 1 if not exactly 1. However, since there doesn't seem to be a positive linear relationship between the two, I would expect the resultant Pearson coefficient to be comparatively weaker and less positive. For x2-x3, I would expect to have a slightly positive Pearson coefficient because of the outliers in the bottom left and top right of the plot, but since Spearman is calculated using rank I don't think it would affect the coefficient calculation as much.

I expect the plot of x1-x3, x1-x4, x2-x4 and x3-x4 to have similar coefficients. x1-x3 is a little noisier but also still has a positive linear relationship (fairly large Pearson coefficient), and is generally monotonically increasing (fairly large Spearman coefficient). x1-x4 does not seem to have a linear relationship, nor does it seem that any monotonicity because points are scattered randomly all over the plot. Consequently, I would think that the Pearson and Spearman coefficients in this case would both be closer to 0. For x2-x4 and x3-x4 as well, the outliers don't seem to create a visible linear trend and sway the Pearson coefficient, and there generally doesn't seem to be any monotonic trend either. I would also think these plots would have Pearson and Spearman coefficients closer to 0.

Table 4: Pearson Correlation Matrix

	x1	x2	x3	x4
x1	1.0000	0.7991	0.7195	-0.0047
x2	0.7991	1.0000	0.5876	-0.0209
x3	0.7195	0.5876	1.0000	-0.0086
x4	-0.0047	-0.0209	-0.0086	1.0000

Table 5: Spearman correlation matrix

	x1	x2	x3	x4
x1	1.0000	0.9719	0.6839	0.0002
x2	0.9719	1.0000	0.6674	-0.0011
x3	0.6839	0.6674	1.0000	-0.0024
x4	0.0002	-0.0011	-0.0024	1.0000

- b. The largest gap found between Pearson and Spearman was in plot x1-x2. The Pearson coefficient 0.7991 whereas the Spearman coefficient was 0.9719, resulting in a gap of 0.1728.
- c. The gap between Pearson and Spearman is a result of differences in how they measure correlation. The x1-x2 plot depicts a monotonic but nonlinear relationship, appearing like an S-shaped curve. As x1 moves from its most negative to most positive values, x2 increases the entire time, but the rate of increase isn't constant like a linear slope would. The curve is steep at the ends but almost flat in the middle. Spearman checks whether the ranks move together, and because the relationship is monotonic with minimal noise, it reports something close to a perfect association ($0.9719 \approx 1$). Pearson instead measures how well a straight line fits the values, so the linear correlation reads meaningfully lower (0.7991) due to the curve. The 0.17 gap is the size of that curvature penalty. It's not noise or randomness pulling Pearson down, but the shape of the function.

I'd argue that the Spearman coefficient is the honest description of x1-x2, since it does successfully reflect that there is a clear strong association between the two variables. Pearson on the other hand, understates the strength of their relationship when correlation is meant to capture how strongly one variable moves with another. Instead, it answers the question of how well does a linear regression model fit x2 as a function of x1. Pearson tells us the strength of a linear relationship between the two variables.

4 Conditional Distributions

Sample Covariance Matrix:

$$\Sigma = \begin{bmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{bmatrix}$$

- a. Given

$$X_1 = a$$

Then

$$X_2 \sim N(\bar{\mu}, \bar{\Sigma})$$

Using $\Sigma_{11} = 1.099875$, $\Sigma_{12} = \Sigma_{21} = 1.696902$, $\Sigma_{22} = 4.104749$, we calculate conditional variance of x_2 given x_1 as:

$$\text{Var}(x_2|x_1) = \bar{\Sigma} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12} = 4.104649 - \frac{1.696902^2}{1.099875} \approx 1.4867$$

Looking at by what factor does the uncertainty of x_2 fall if you learn the value of x_1 , given the values in the partitioned matrix are scalars:

$$\frac{\text{Var}(x_2|x_1)}{\text{Var}(x_2)} = \frac{\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}} = 1 - \frac{\text{cov}(x_1, x_2)^2}{\text{var}(x_1)\text{var}(x_2)} = 1 - \left(\frac{\text{cov}(x_1, x_2)}{\sigma_1\sigma_2}\right)^2 = 1 - \rho^2$$

In the context of this data, learning the value of x_1 shrinks our uncertainty about x_2 by a factor of

$$1 - \frac{\text{cov}(x_1, x_2)^2}{\text{var}(x_1)\text{var}(x_2)} = 1 - \frac{1.696902^2}{1.099875 \cdot 4.104749} = 0.3622$$

So learning x_1 shrinks the uncertainty of x_2 down to about 36.22% of its unconditional variance.

- b. No, the factor $1 - \rho^2$ does not depend on the value of x_1 observed:

$$1 - \rho^2 = 1 - \frac{\text{cov}(x_1, x_2)^2}{\text{var}(x_1)\text{var}(x_2)}$$

which does not contain any a term. Consequently, the factor by which the variance of x_2 changes knowing x_1 is not affected by the given value of x_1 .

- c. The conditional mean of x_2 given x_1 :

$$\bar{\mu} = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(a - \mu_1)$$

Given the values in the partitioned matrix are scalars, it can be rewritten as:

$$\bar{\mu} = \mu_2 + \frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)}(x_1 - \mu_1)$$

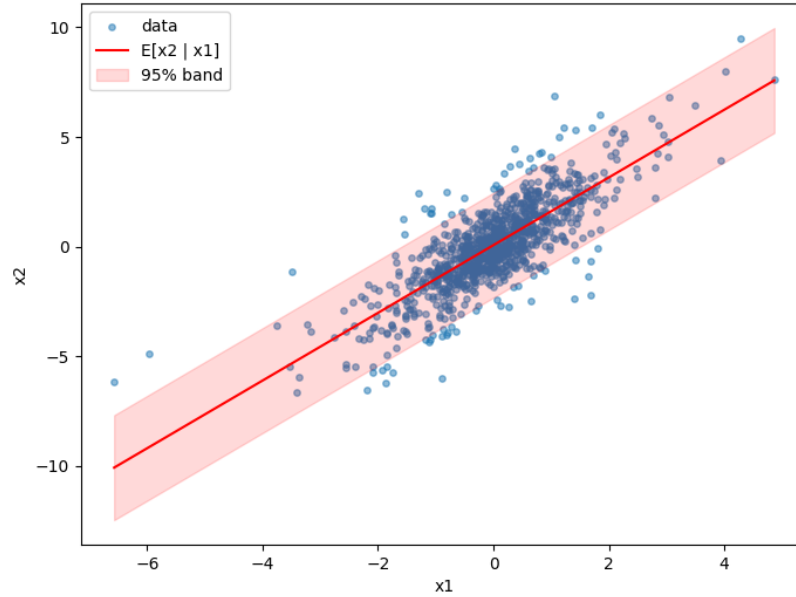
The coefficient on x_1 , $\frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)}$, is just the estimated OLS coefficient, β . This means that the expression can be rearranged as an OLS regression equation, where:

$$\bar{\mu} = E[x_2|x_1] = \alpha + \beta x_1$$

where

$$\alpha = \mu_2 - \beta\mu_1$$

Figure 3: Plot of Conditional Expectation



The expression for conditional mean of x_2 given x_1 is: $0.0711 + 1.5428x_1$.

- d. Fraction of observations inside the 95% band: 0.9350

Table 6: Split Sample and its Coverage

Region	# Observations	Coverage
$ z_1 < 1$	759	0.9565
$1 \leq z_1 < 2$	190	0.8579
$ z_1 \geq 2$	51	0.9020

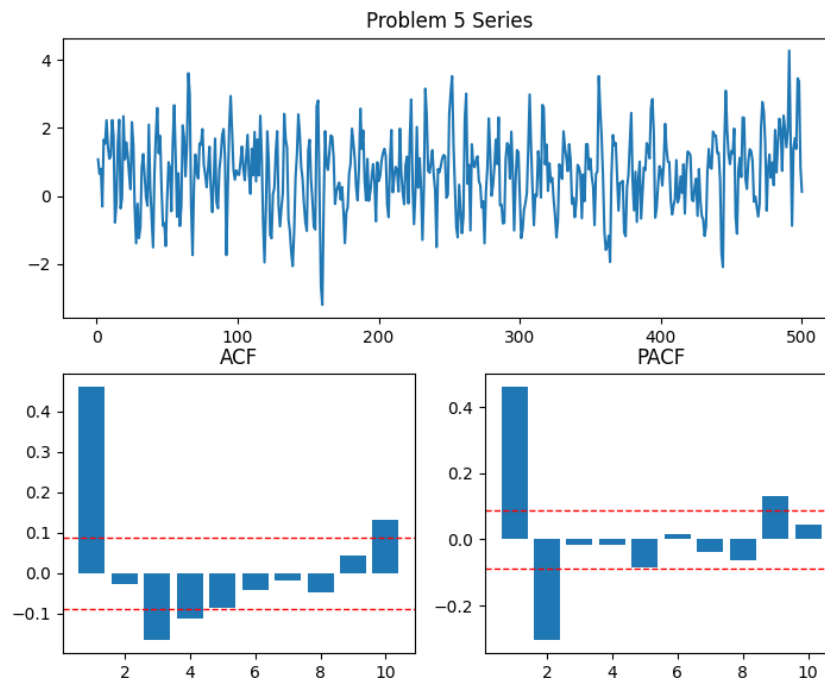
e.

- f. Given the coverage is not flat across the buckets (coverage was 0.9565 near the mean, dropped to 0.8579 in the 1-2 standard deviation bucket, and 0.9020 beyond 2 standard deviations), that tells us that the assumption made in part b) regarding the conditional variance, that $\text{var}(x_2|x_1)$ does not change as x_1 changes is incorrect. Clearly, $\text{var}(x_2|x_1)$ varies across the range of x_1 . The real spread of x_2 around the conditional mean changes depending on how far x_1 is from its own mean. Σ is forced to average over regions with different spreads.

What survives from part c) is the conditional mean. The regression line still works because $\beta = \frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)}$ doesn't require homoskedasticity to be an unbiased estimate. What does not survive is the constant width band built around the regression line, since the true conditional variance changes with x_1 .

5 Identifying an AR or MA Order

Figure 4: Plot of Series, its ACF, and its PACF



- a. The PACF is cut off whereas the ACF decays. This suggests that does is $AR(p)$ process with order $p = 2$ since the PACF is cut off after lag 2.
- b. The rule I used was that a lag is treated as statistically significant if the sample ACF or PACF value at that lag falls outside the $\pm 1.96/\sqrt{n}$ region. I use the same significance band as shown in class, giving a 95% confidence threshold. A function "cuts off after lag p " when all the bars (for the most part) stay inside the region for every lag beyond p , while a function that keeps producing increasingly smaller bars near or outside the band as the lag increases is read as "decaying" rather than cutting off.
- c. After fitting the various $AR(p)$ and $MA(q)$ models, I got the following AICc values:

Table 7: AICc for Various Models

Model	AICc
AR(1)	1418.5870
AR(2)	1372.4383
AR(3)	1374.3432
MA(1)	1389.0843
MA(2)	1381.2316
MA(3)	1374.1165

- d. The AICc-selected model is AR(2) given that it has the lowest AICc value. This matched my prediction made using Table 3 in the Week 2 lecture notes.
- e.

$$AR(2) : y_t = 0.4704 + 0.6012y_{t-1} - 0.3026y_{t-2} + \epsilon_t, \hat{\sigma}^2 = 0.8950$$

$$AR(3) : y_t = 0.4780 + 0.5962y_{t-1} - 0.2927y_{t-2} - 0.0165y_{t-3} + \epsilon_t, \hat{\sigma}^2 = 0.89556$$

$AR(3)$ nests $AR(2)$. The two models are nearly identical everywhere they have in common: the constant and the first two AR coefficients are about the same. The third coefficient, -0.0165, is small and thereby does not contribute much. Variance also barely changed after adding the third parameter. These further suggest that $AR(2)$ is the best model.

AICc is defined as:

$$AICc = -2\ln(\hat{L}) + 2k + \frac{2k^2 + 2k}{n - k - 1}$$

When adding the third parameter, variance went up by a negligible amount, from 0.8950 to 0.89556. While this decreased the $-2\ln(\hat{L})$ term, it was not enough to offset the increase in $2k + \frac{2k^2 + 2k}{n - k - 1}$ as a result of the now higher number of parameters. This ultimately increased AICc, making $AR(3)$ a less attractive candidate than $AR(2)$.

R^2 tends to reward models with a higher numbers of parameters the model. This is because $R^2 = 1 - SSE/SST$, and $AR(3)$'s SSE can never be worse than $AR(2)$'s since it can set the coefficient on y_{t-3} to be zero. Consequently, R^2 would think that $AR(3)$ would be a better choice than $AR(2)$. AICc on the other hand, penalizes additional parameters if it does not bring about a larger change to log-likelihood. In this case it did, hence why they would have not made the same choice.